

Grade 7
Unit 10
Lesson 7 Practice

Name Answer Key
Date _____
Period _____

1. Solve the equation $4(0.5p + 1.5) = 12$ using your preferred method.
 $4(0.5p + 1.5) = 12$

$$\begin{array}{r} \frac{4}{0.5p + 1.5} = \frac{4}{3} \\ \underline{-1.5 \quad -1.5} \\ 0.5p = 1.5 \\ \underline{0.5 \quad 0.5} \\ p = 3 \end{array}$$

2. Use substitution to verify your solution in question 1.

$$\begin{array}{l} 4(0.5(3) + 1.5) = 12 \\ 4(1.5 + 1.5) = 12 \\ 4(3) = 12 \\ 12 = 12 \end{array}$$

3. Solve the equation $\frac{5}{2} = \frac{1}{4}(z - 8)$ using your preferred method.

$$\begin{array}{l} 4 \cdot \frac{5}{2} = 4 \cdot \frac{1}{4}(z - 8) \\ 10 = z - 8 \\ \underline{+8 \quad +8} \\ 18 = z \end{array}$$

4. Use substitution to verify your solution in question 3.

$$\begin{array}{l} \frac{5}{2} = \frac{1}{4}(18 - 8) \\ \frac{5}{2} = \frac{1}{4}(10) \\ \frac{5}{2} = \frac{5}{2} \end{array}$$

5. Solve the equation $18 = \frac{3}{4}(36.4 - n)$ using your preferred method.

$$\begin{array}{l} \frac{4}{3} \cdot 18 = \frac{4}{3} \cdot \frac{3}{4}(36.4 - n) \\ 24 = 36.4 - n \\ \underline{-36.4 \quad -36.4} \\ -12.4 = -n \\ \underline{-1 \quad -1} \\ 12.4 = n \end{array}$$

6. Use substitution to verify your solution in question 5.

$$\begin{array}{l} 18 = \frac{3}{4}(36.4 - 12.4) \\ 18 = \frac{3}{4}(24) \\ 18 = 18 \end{array}$$

7. The perimeter of a regular pentagon with equal sides is 12 inches. If the length of each side is $w + \frac{2}{5}$, the equation $5\left(w + \frac{2}{5}\right) = 12$ can be used to find the value of w . Determine the value of w .

$$\begin{array}{rcl} 5\left(w + \frac{2}{5}\right) & = & 12 \\ 5w + 2 & = & 12 \\ \underline{-2 \quad -2} & & \\ 5w & = & 10 \end{array} \qquad \begin{array}{rcl} 5w & = & 10 \\ \underline{5} \quad \underline{5} & & \\ w & = & 2 \end{array}$$

8. The length of each side of the regular pentagon is $2\frac{2}{5}$ inches because $12 \div 5 = 2\frac{2}{5}$. Explain how the solution to the equation $5\left(w + \frac{2}{5}\right) = 12$ can be used to show this as well.

If the length of one side is represented by $w + \frac{2}{5}$, and w is equal to 2, then $2 + \frac{2}{5} = 2\frac{2}{5}$ and this is one side of the pentagon. Multiply this by 5 to get a perimeter of 12.

Sade was asked to solve the equation $\frac{2}{3}(b + 2.5) = 9.5$. Her work is shown.

$$\begin{array}{rcl} \frac{2}{3}(b + 2.5) & = & 9.5 \\ \frac{2}{3} \cdot b + \frac{2}{3} \cdot 2.5 & = & 9.5 \cdot \frac{2}{3} \\ \frac{2}{3}b + \frac{5}{3} & = & \frac{19}{3} \\ \frac{2}{3}b + \frac{5}{3} - \frac{5}{3} & = & \frac{19}{3} - \frac{5}{3} \\ \frac{2}{3}b & = & \frac{14}{3} \\ \frac{3}{2} \cdot \frac{2}{3}b & = & \frac{3}{2} \cdot \frac{14}{3} \\ b & = & \frac{14}{2} = 7 \end{array}$$

9. Explain where Sade made a mistake.
Sade multiplied both sides of the equation by $\frac{2}{3}$ when she should have distributed $\frac{2}{3}$ to $(b + 2.5)$.
10. Determine the correct solution to the equation.

$$\begin{array}{rcl} \frac{2}{3}(b + 2.5) & = & 9.5 \\ \frac{2}{3}b + \frac{5}{3} & = & 9.5 \\ \underline{-\frac{5}{3} \quad -\frac{5}{3}} & & \\ \frac{3}{2} \cdot \frac{2}{3}b & = & \frac{47}{6} \cdot \frac{3}{2} \\ b & = & 11.75 \end{array}$$